

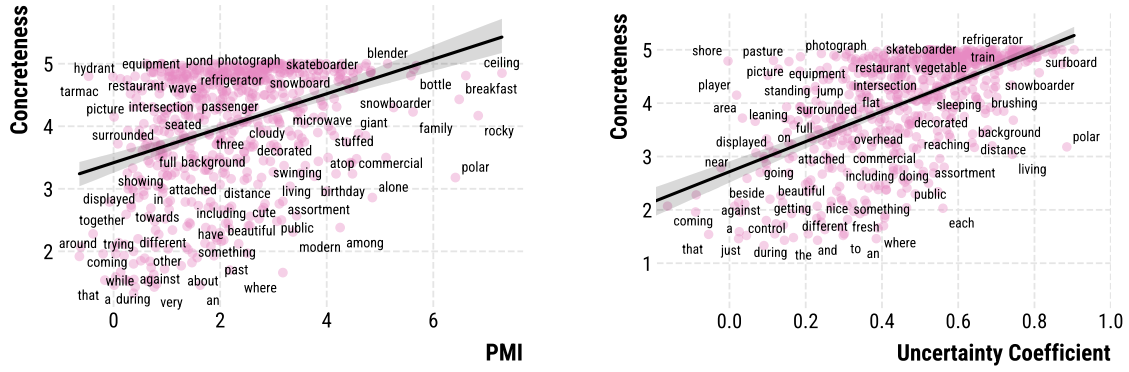


Figure 4: Correlation between human concreteness ratings and type-level groundedness (PMI; left, $\rho = 0.368$) or uncertainty coefficient (right, $\rho = 0.609$): i.e., the average ratio between LM surprisal and captioning model surprisal.

MI estimates For the language-level MI estimates in Figure 1, we consider the separate effects of language, dataset, and POS on groundedness. Because the meanings (images) are matched across languages, this allows us to estimate and control for some languages having consistently larger or smaller MI estimates (due to language-specific variation in our neural estimators). We find significant effects of all 3 factors, but they differ dramatically in how much variation they explain. The effect of dataset is extremely small, explaining 0.5% of the observed variance ($F_{3,816} = 5.71$, $p < 0.01$). Language identity has a larger effect, explaining 8.2% of the variance ($F_{29,789} = 6.42$, $p < 0.001$). However, word class dominates, explaining most of the total variance (57.3%, $F_{12,806} = 775$, $p < 0.001$), and 62.8% of the remaining variance after controlling for variance due to dataset and language. Altogether, these factors explain 65.6% of the variance, leaving the remaining variance to cross-linguistic differences in the MI of specific parts of speech.

PMI distributions We also investigate how much variation in the full distribution of contextual groundedness estimates (PMIs) is explained by word class (shown in Figure 3). Within a POS, groundedness is expected to vary substantially: for example, some (concrete, visually distinct) nouns have much higher PMI with the image than others, and tokens of the same word type also have different groundedness (e.g. “lot” referring to a location vs. “lot” as a quantity expression). Therefore, we expect word class to explain much less variance than in the overall MI estimates. Language, dataset, and their interaction account for 2.4% of the total variation in PMIs across the three datasets ($F_{64,107} = 4727$, $p < 0.001$). Word class accounts for 12.0% of the total variation

($F_{12,107} = 123583$, $p < 0.001$). Additionally, the interaction between word class and language (cross-linguistic variation in the means of word classes) accounts for only an additional 1.6% of the total variation ($F_{330,107} = 602.5$, $p < 0.001$), despite having many degrees of freedom. So cross-linguistically consistent tendencies comprise the bulk of the explainable variance in the overall PMI distribution across these three datasets—5 times as much as language and dataset, and 7.5 times as much as language differences in POS groundedness.

5.4 Semantic dimension of the measure

In this section we explore the semantic properties of the groundedness measure introduced here, comparing it to semantic norms related to contentfulness that are widely used in psycholinguistics. One potential advantage of our method is the ease with which it allows the rating of individual word tokens in context; however, existing ratings tend to be for words in isolation (word types). We focus our analysis here on English and on word types which occur at least 30 times in the COCO(-35L) validation set, averaging across occurrences to obtain an estimate of the average type-level groundedness.

We compare to three different psycholinguistic norms: imageability, concreteness, and strength of visual experience. Such norms are measured by providing a definition and examples of low- and high-value words to raters, who then rate words on a Likert Scale. For imageability, we use the Glasgow Psycholinguistic Norms (Scott et al., 2019). For concreteness, we use the Brysbaert et al. (2014) norms. For strength of visual experience, we use the Lancaster Sensorimotor Norms (Lynott et al., 2020). Results for concreteness are shown in Figure 4 (left). We observe fairly weak (though signif-

icant, $p < 0.001$) correlations with groundedness using Spearman’s ρ (Imageability: $\rho = 0.288$, Concreteness: $\rho = 0.368$, Visual strength: $\rho = 0.212$).

We find these weak correlations are partly due to the *informativity* aspect of our measures, which seems not to play as large of a role in human ratings (e.g. woman is just as concrete as skateboard, but less informative and also less grounded by our measure). To account for differences in baseline (LM) word informativity, we can normalize the PMI scores by the LM surprisal, yielding the uncertainty coefficient (Theil, 1970): the proportion of the LM surprisal explained by the PMI. Regressing this value against the psycholinguistic norms, stronger correlations emerge (Imageability: $\rho = 0.548$, Concreteness: $\rho = 0.609$ as shown in Figure 4 (right), Visual strength: $\rho = 0.320$). This suggests that the differences between groundedness and surprisal are associated with concreteness. However, this measure collapses differences between word classes in overall informativity/surprisal.

In some cases, outliers are due to contextual effects. For example, in our data the word “polar” (high groundedness, moderate concreteness) occurs exclusively as the first word in the multi-word expression “polar bear” which is highly concrete, imageable, and visual; while ratings based on the word type are for the more abstract geographical concept. Other words with divergent scores between human-based and model-based methods tend to be those which frequently occur in contexts where they are highly expected (e.g. “shore” which tends to occur in limited syntactic contexts and after the appearance of words like “boat,” “lake,” or “surfers”), or words which are often used non-specifically in the image captioning context (e.g. “photo” exhibits very low PMIs, because captions frequently begin with “A photo of ...”).

6 Discussion and Conclusion

We have proposed a grounded approach to typology, using images as a proxy for sentence meaning. Using information theory and neural models, we define *groundedness*, a measure of a token’s association with the meaning expressed in a sentence. Our results demonstrate that word classes display consistent patterns in terms of their groundedness across a typologically diverse sample of languages. We find these patterns can be described as a continuous cline which generalizes the traditionally dichotomous distinction between lexical and func-

tional word classes into a gradient one. However, our results suggest grammatical word classes still carry semantic content. We find that nouns > adjectives > verbs, in line with a view of these classes as a continuum; yet, our results contradict claims that adpositions are more lexical than other functional classes. Our measure is related to surprisal, but diverges from it, particularly for concrete words.

While this work has focused on word classes, groundedness enables the exploration of other aspects of how languages express function through form. Future work could investigate in detail under what conditions “functional” items have higher groundedness. For example, do more spatial adpositions and determiners have higher groundedness than less spatial ones? Humans tend to have difficulty scoring highly abstract and grammaticalized words, and getting contextual scores is difficult with existing psycholinguistic approaches: groundedness opens new ways to address these questions.

Our approach is also suitable for studying non-prototypical word class organizations, such as languages which do not clearly distinguish between adjectives and verbs (Korean; Maling and Kim, 1998), or languages that split individual word classes into distinct sub-classes (Japanese adjectives; Backhouse, 1984). Future work should look at both formal and semantic sub-classes of parts of speech—such as gerunds, participles, and different semantic classes of verbs (as in VerbNet; Kipper Schuler et al., 2009)—investigating their groundedness and how it aligns with or varies from existing metrics. In particular, we conjecture that boundary classes (e.g. gerunds) may display intermediate groundedness (between nouns and verbs) compared to prototypical members of those classes. Groundedness makes it possible to test this conjecture with reference to the contexts in which words appear, which is needed for distinguishing syncretic forms.

Our approach can also cover any classes which can be defined over linguistic units, such as morphemes, phrases, or semantic classes. For instance, future work could explore the claim that inflections are more “grammatical” than derivations (Booij, 2007; Haley et al., 2024). Similarly, our measure could be used to study the lexicalization or grammaticalization of constructions (as a decrease in groundedness over time). To support such work, we release our groundedness scores online.⁶

Going beyond the details of the approach here,

⁶<https://osf.io/bdhna/>

our work generally suggests a role for multimodal models in computational typology similar to the one played by language models in the past decade (e.g. Pimentel et al., 2023; Cotterell et al., 2018; Ackerman and Malouf, 2013). While language coverage remains more limited than text models, the latest multimodal models and datasets cover enough typologically and culturally diverse languages to make them worth studying—and we anticipate coverage will only improve. Further, the ability of multimodal models to provide an empirically grounded (if imperfect) representation of meaning makes them uniquely valuable for quantitatively addressing questions about the relation between form and function in language. Our work provides the first study of this kind, and we hope that by demonstrating the utility of this approach and releasing our groundedness scores we will inspire other researchers to follow suit.

7 Limitations

Our approach has a number of important limitations. These limitations should inform the interpretation of results here, as well as any future studies considering using these techniques.

First, our operationalisation of meaning as an image is necessarily a simplification and has numerous implications for our results. Notably, the choice of images rather than videos (motivated by model quality and availability) as the representation of meaning has major implications for verbs, which tend to have meanings which are more temporally extended. This choice also has substantial implications about the variety of language which can be analyzed—many types of language use, such as metaphoric extension, are likely to be much less frequent in image captions than in other domains of language use: such phenomena are perhaps best studied using a different technique. This problem is compounded by the fact that existing multilingual corpora for these datasets remain fairly small—thus the analysis of long-tail phenomena in language using these methods is likely not yet possible.

Compared to existing methods in typology, this method trades human effort for computational resources. While we make both our models and data available, significantly lessening the burden on future studies, the models here contain between two and three billion parameters, and the image models have very long sequence lengths due to the image tokens. Inference on new data is therefore fairly

expensive with current technologies.

Further, there remain significant limitations on the languages which can be studied with these approaches. Currently available models cover just 16 languages which are not part of the Indo-European language family, and entire areal typological regions like the Americas are not covered. We hope that the quality and coverage of these models can continue to improve, and that findings based on current models can be revisited and replicated with newer models.

Finally, we rely on automatic part of speech tagging based on Universal Dependencies for the analyses here (see Appendix A for further information). Overall, the accuracy of the Stanza tagger is high for the Universal Dependencies corpora of the languages studied here (96% on average); however, it is not uniformly accurate across languages. Vietnamese has the lowest average accuracy, with 81.5% on their test set; however, our data is different in domain from many of the universal dependencies corpora, so the accuracy might be somewhat lower or higher. Universal Dependencies part of speech tags are not entirely without controversy as well—for instance, some linguists would argue that Korean does not have an adjective class, but UD uses one. It is possible that choices or inconsistencies in the assignment of POS tags according to UD could impact some MI estimates. In summary, noise due to POS tagging may have some influence on the results here, but is unlikely to affect our main conclusions.

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